Statistics MCQs and Theory Questions

Multiple Choice Questions (MCQs)

1. In a normal distribution, what percentage of data lies within two standard deviations of the mean ($\mu \pm 2\sigma$)?
a) 68%
b) 95%
c) 99.7%
d) 100%

2. Which Excel function is used to calculate the probability of a value less than or equal to x in a normal distribution?
a) NORM.INV
b) NORM.S.DIST
c) NORM.DIST with cumulative = TRUE
d) NORM.DIST with cumulative = FALSE

3. The Poisson distribution is characterized by:
a) Two parameters: n and p
b) A constant probability of success per trial
c) A single parameter λ (mean = variance)
d) A symmetric bell-shaped curve

4. In a Poisson distribution with $\lambda = 4$, what does λ represent?
a) The variance of the distribution
b) The average number of events in the interval
c) The number of trials
d) The probability of success

5. The odds ratio is calculated as:
a) Probability of event / Probability of non-event
b) (Odds in exposed group) / (Odds in unexposed group)
c) (Probability in exposed) – (Probability in unexposed)
d) Standard deviation / Mean

6. In a 2×2 contingency table for a case-control study, the odds ratio is:
a) (a×d) / (b×c)
b) (a×b) / (c×d)
c) (a+c) / (b+d)
d) (a+b) / (c+d)

7. The interquartile range (IQR) is:
a) The range between the minimum and maximum values
b) The difference between Q3 and Q1
c) The median of the dataset
d) The standard deviation multiplied by 2

8. In a box and whisker plot, the upper whisker typically extends to:
a) The maximum value in the dataset
b) Q3 + 1.5 × IQR or the maximum value, whichever is smaller
c) Q3 + 3 × IQR
d) The mean + standard deviation

9. Listwise deletion of missing values means:
a) Deleting missing values only from one variable
b) Deleting any row that contains at least one missing value
c) Replacing missing values with the mean
d) Deleting columns with more than 50% missing values

10. KNN imputation works by:
a) Replacing missing values with the global mean
b) Finding similar cases (neighbors) and averaging their values
c) Deleting all cases with missing data
d) Replacing missing values with zeros

Set - B (Theory Questions)

1. Explain the 68-95-99.7 empirical rule for a normal distribution. How does this rule help in understanding data variability?

2. What is an interquartile range (IQR)? Explain how the 1.5 × IQR rule is used to identify outliers in a dataset and how this is visually represented in a box and whisker plot.

3. Distinguish between positive correlation, negative correlation, and no correlation. Give one real-world example of each from an engineering or manufacturing context.

4. What is simple linear regression? Explain the terms dependent variable, independent variable, slope, and intercept using the equation $Y = mX + b$.

5. What is KNN imputation? How does it work to handle missing values in a dataset? State one advantage and one disadvantage of using KNN over mean imputation.